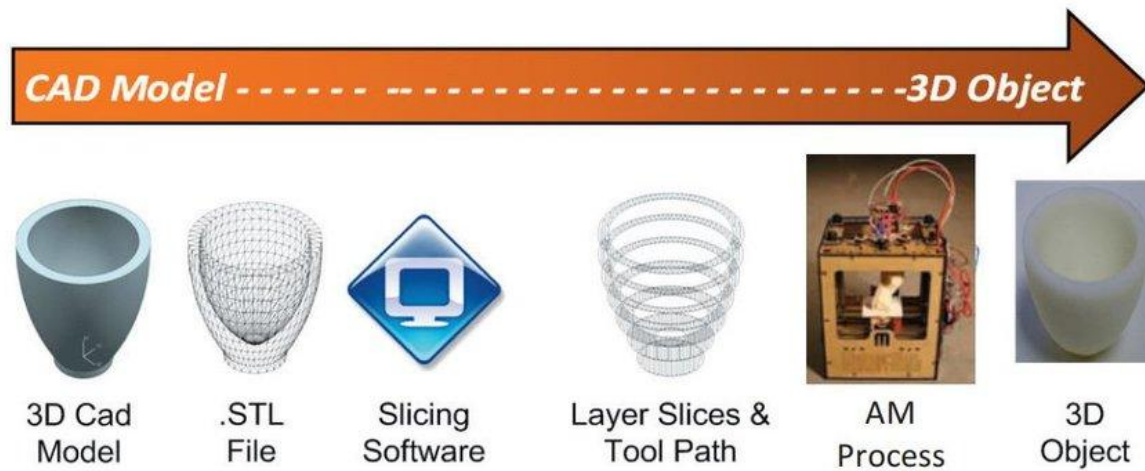


INTRODUCTION TO ADDITIVE MANUFACTURING



COURSE CONTENT

- 1.1.1 Introduction
- 1.1.2 Additive Manufacturing
- 1.1.3 Steps in Additive Manufacturing
- 1.1.4 Classification of AM Processes
- 1.1.5 AM in New Product Development
- 1.1.6 Materials used in AM
- 1.1.7 Additive v/s Conventional Manufacturing processes
- 1.1.8 Applications of AM in Various Domains

Prof. Shiva Krishna J

Dept. of Mechanical & Automation Engg., Agnel Institute of Technology and Design, Assagao

1.1.1 INTRODUCTION

A prototype is a first or preliminary version of a device, system, or vehicle from which subsequent forms are developed or refined.

Unlike the conventional view that a prototype must be a physical object, the term broadly covers all types of representations used in the product development process. These can include mathematical models, pencil sketches, virtual models, foam models, 3D-printed parts, or fully functional physical approximations of the product.

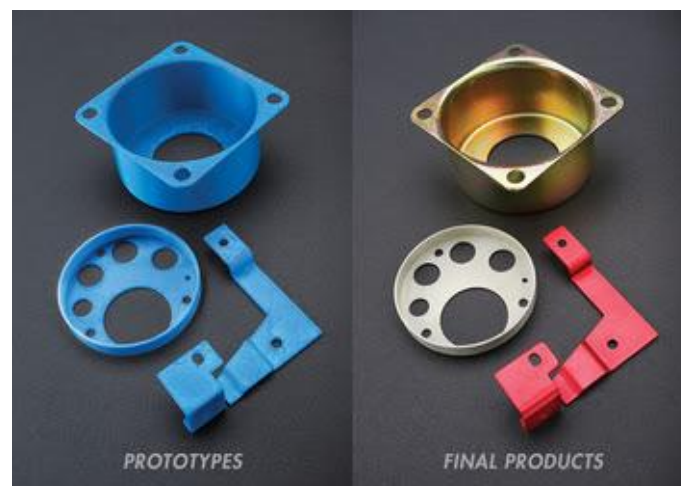
Prototyping is the process of creating these representations. It can range from running a simulation on a computer program to building a working physical model that mirrors the final product.

ROLE OF PROTOTYPES IN PRODUCT DEVELOPMENT

Prototypes play multiple critical roles in product design and development, such as:

1. Experimentation and Learning: Exploring design ideas, testing feasibility, and identifying potential flaws at an early stage.
2. Testing and Proofing: Validating performance, safety, ergonomics, and durability before large-scale production.
3. Communication and Interaction: Helping designers, engineers, clients, and stakeholders visualize the product and provide constructive feedback.
4. Synthesis and Integration: Bringing together mechanical, electrical, software, and user-interface aspects into a unified model.
5. Scheduling and Milestones: Serving as tangible checkpoints in the development timeline to assess progress and readiness for subsequent phases.
6. Risk Reduction: Minimizing the cost and impact of errors by identifying and resolving issues before mass production.
7. Innovation Support: Allowing room for iterative improvements and encouraging creative exploration.

PROTOTYPE v/s FINAL PRODUCT



Difference between a prototype and Final Product

Although prototypes serve as the foundation for the final product, the creation of a prototype differs in several fundamental ways:

1. Material:

- Final products are often made with durable, high-quality, and sometimes costly materials.
- Prototypes, on the other hand, may use substitute or easily machinable materials to reduce costs and speed up development.
- In some cases, final materials may not even be available yet during the prototyping stage.

2. Process:

- Final products typically use mass-production processes (e.g., injection molding, die casting, automated assembly).
- Prototypes are usually fabricated using rapid or low-volume methods such as CNC machining, 3D printing, laser cutting, or handcrafting.
- This often results in visual or structural differences between the prototype and the final product.

3. Verification and Testing:

- Final products undergo rigorous quality assurance processes, including statistical sampling, regulatory compliance checks, and durability testing.
- Prototypes, however, are usually inspected individually, with the expectation of modifications, rework, or redesign. Some compliance requirements may be waived during this stage.

4. Cost and Time:

- The final product aims for cost-effectiveness in large-scale manufacturing.
- Prototypes may be more expensive per unit due to specialized fabrication and one-off tooling, but they save overall costs by preventing large-scale errors.

5. Functionality:

- Prototypes may be partial or full representations of the final product.
- Types include visual prototypes (appearance only), functional prototypes (performance testing), and concept prototypes (early idea validation).
- The final product integrates all required functionalities with consistency, reliability, and scalability.

Engineers and prototype specialists strive to minimize the gap between a prototype and the final product. For example:

- If the same material cannot be used, substitutes with similar physical or mechanical properties are chosen.
- If mass-production tooling is unavailable, rapid prototyping technologies simulate the required precision and performance.
- Iterative prototyping cycles help progressively refine the product until it closely matches the intended final design.

PHASES OF PROTOTYPING

- Prototyping or model making is an age-old practice, rooted in the need to transform conceptual designs into tangible forms. A prototype serves as a bridge between design and production, enabling visualization, testing, and refinement before mass manufacturing.
- Traditionally, prototypes have been fabricated through processes such as material removal, casting, moulding, and adhesive joining, using a variety of materials including aluminium, zinc, urethanes, plastics, and wood.
- Over time, prototyping has evolved through three distinct phases, the latter two driven by advancements in digital technology and automation. Much like the evolution of computer graphics, physical prototyping has now entered its third and most advanced phase.

1. First Phase: Manual Prototyping:

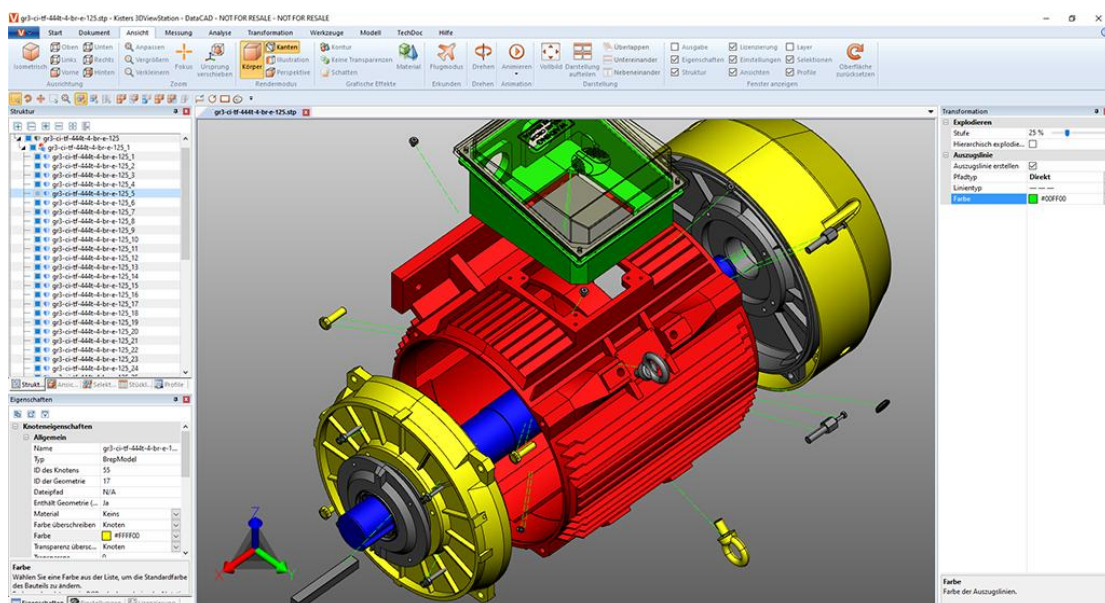
- Manual prototyping began centuries ago as humans crafted tools and products to meet their needs.
- In this phase, prototypes were often simple, handmade, and craft-based, relying heavily on the skill of artisans.
- Fabrication was time-consuming and labour-intensive, typically taking around four weeks for a moderately complex prototype.
- Techniques included hand carving, wood/metal shaping, and basic casting methods.

Limitations:

- Low accuracy and repeatability.
- Difficulties in scaling up to mass production.
- Long fabrication cycles.

2. Second Phase: Soft (or) Virtual Prototyping:

- With the rise of CAD (Computer-Aided Design), CAE (Computer-Aided Engineering), and CAM (Computer-Aided Manufacturing) in the early 1980s, virtual prototyping emerged.



Virtual Prototyping

- Designers could now create digital models, simulate stress, strain, thermal effects, and other behaviours by specifying precise material properties.
- Fabrication of physical models still relied on traditional/manual methods, taking up to 16 weeks despite improvements like CNC machining.

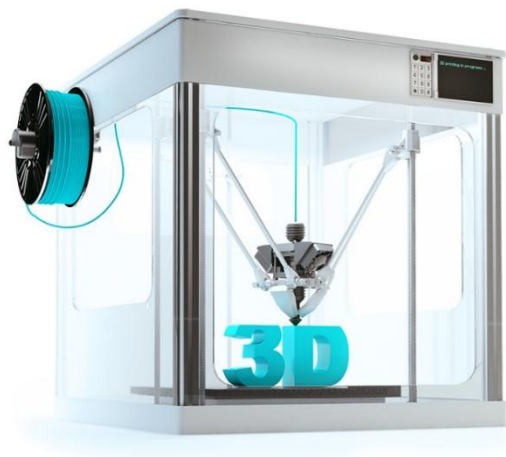
Advantages:

- Multiple design iterations could be performed quickly by altering parameters in the software.
- Complex designs nearly twice as complex as before became feasible.
- Allowed early detection of flaws without physical fabrication.

Limitations:

- Virtual prototypes cannot anticipate or simulate all real-world phenomena. Unmodeled effects may cause performance gaps in actual usage.

3. Third Phase: Rapid Prototyping:



Rapid Prototyping

- The late 1980s marked a revolutionary leap with the advent of Rapid Prototyping (RP), also known as Solid Freeform Fabrication (SFF), Layer Manufacturing Technology, Additive Manufacturing (AM) and Desktop Manufacturing
- Uses layer-by-layer fabrication directly from CAD models.
- Eliminates tooling, moulds, or machining setups, drastically reducing lead time.
- Capable of producing prototypes with 3× the complexity of 1970s designs in as little as 3 weeks.
- Since 1988, more than 20 different RP techniques have been developed (e.g., Stereolithography, Selective Laser Sintering, Fused Deposition Modelling, Laminated Object Manufacturing).

Advantages:

- Rapid turnaround for complex parts.
- High accuracy and repeatability.
- Supports a wide range of materials (plastics, metals, ceramics, composites).
- Enables functional testing (fit, form, performance).

1.1.2 ADDITIVE MANUFACTURING

Additive Manufacturing (AM), commonly known as 3D Printing, is the layer-by-layer fabrication of three-dimensional physical models directly from CAD data.

- It represents a group of advanced techniques that enable the rapid and accurate creation of parts, assemblies, or functional prototypes directly from digital models.
- Unlike traditional subtractive or formative manufacturing methods, AM builds objects by adding material only where needed, making it an efficient, flexible, and innovative approach to product development and production.

KEY FEATURES AND ADVANTAGES OF ADDITIVE MANUFACTURING

1. Layer-by-Layer Fabrication:

- AM constructs objects by stacking thin cross-sections generated from CAD data.
- This approach allows the creation of complex geometries, including lattice structures, internal cavities, thin walls, and *parts within parts*, which are extremely difficult or impossible to achieve using conventional methods.

2. Simplified Construction Process:

- A 3D model is digitally sliced into layers, which are then printed consecutively.
- Example: Instead of machining a sphere, AM builds it through successive layers of circular slices, achieving precision with minimal effort.

3. Reduced Human Intervention:

- Once the CAD design is prepared and the machine is set up, AM systems usually operate unattended until the build is complete.
- Post-processing (e.g., support removal, surface finishing, curing) is required, but operator involvement is much lower than in conventional manufacturing.

4. Time and Cost Efficiency:

- Prototypes and functional parts can be produced in hours instead of weeks.
- Eliminates the need for expensive tooling, moulds, and complex setups.
- Materials used in AM are often more economical, especially for prototyping and low-volume production.

5. Accessibility and Flexibility:

- AM machines range from desktop-scale printers (suitable for offices and classrooms) to industrial-grade systems for aerospace, automotive, and biomedical industries.
- Designers can quickly modify and reprint parts, speeding up the design–test–iterate cycle.

6. Functional Prototyping and End-Use Parts:

- AM produces functional prototypes that can be tested for fit, form, and performance.
- Increasingly, AM is being used to manufacture end-use parts, such as lightweight aerospace components, medical implants, and customized consumer products.

7. Enables Innovation:

- Provides unparalleled design freedom by removing traditional manufacturing constraints.
- Supports topology optimization, lattice structures, and multi-material printing.
- Encourages sustainability by reducing material waste and enabling localized, on-demand production.

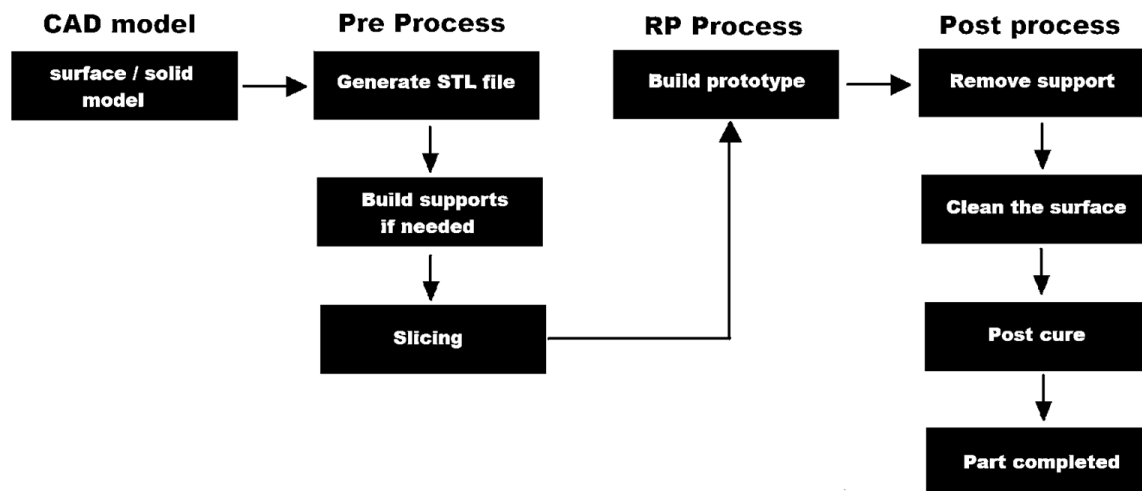
COMMONLY USED TERMS FOR ADDITIVE MANUFACTURING

Additive Manufacturing is also referred to by several related terms:

- 3D Printing
- Direct CAD Manufacturing
- Desktop Manufacturing
- Instant Manufacturing
- CAD-Oriented Manufacturing
- Layer Manufacturing
- Material Deposition Manufacturing
- Material Addition Manufacturing
- Solid Freeform Fabrication (SFF)

1.1.3 STEPS IN ADDITIVE MANUFACTURING

AM in its basic form can be described as the production of 3D parts from CAD data within a reduced time scale. The basic methodology of all AM processes can be summarized as shown in the following steps:



Steps in Additive Manufacturing

1. Development of a CAD Model:

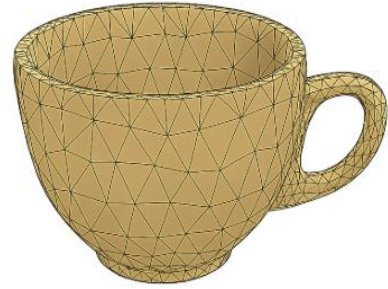
The process begins with the generation of a CAD model of the desired object, which can be done by one of the following ways:

- Conversion of an existing two-dimensional (2D) drawing
- Importing scanned point data into a CAD package
- Creating a new part in CAD using various solid modelling packages
- Altering an existing CAD model



2. Generation of STL File:

The developed 3D CAD model is tessellated and converted into Standard Triangulation Language (STL) files that are required for AM processes. Tessellation is a piecewise approximation of surfaces of a 3D CAD model using a series of triangles. The STL file represents the surface of the model as an array of triangles and consists of the X, Y, and Z coordinates of the three vertices of each triangle, along with an index that describes the orientation of the surface normal.



3. Build Support Structures:

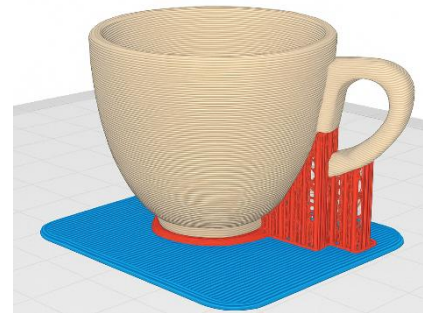
Since parts are built layer by layer, some areas may be unsupported or overhanging, which could cause distortion. Therefore, certain AM processes require support structures or a base to be added to the file. These supports are fabricated along with the model and later removed.

4. Slicing:

Slicing involves creating contours of the geometry at various heights according to the chosen layer thickness. Once the STL file has been generated from the original CAD data, the next step is to slice the object to create a slice file (SLI).

- Thicker slices → faster build, but rougher surface finish.
- Thinner slices → slower build, but finer detail.

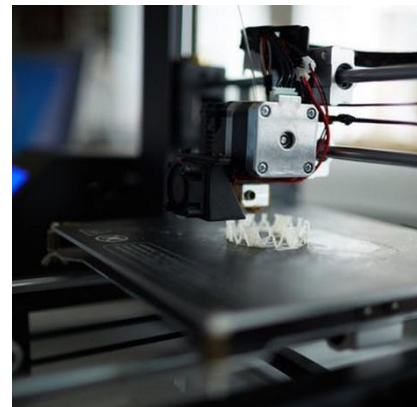
The sliced files are then merged into a final build file.



5. Building the Part:

The AM process is additive, i.e., the part is built layer by layer from the bottom up. Each new layer is automatically bonded to the previous one. Depending on the material and technology, bonding can be achieved through:

- Ultraviolet (UV) light
- Carbon dioxide lasers
- Heat-sensitive adhesives
- Melting and solidification of the material itself



6. Post-Processing:

After the build, the part is removed from the machine and subjected to post-processing. This may include:

- Removing support structures
- Curing or sintering for improved strength
- Filling voids
- Surface finishing (sanding, polishing, coating, painting)

The level of post-processing depends on the final application and quality requirements of the part.

1.1.4 CLASSIFICATION OF ADDITIVE MANUFACTURING PROCESSES

AM processes can be classified in multiple ways depending on the initial form of the build material and the underlying process mechanism.

I. BASED ON BUILD MATERIALS:

1. Liquid-based Systems:

- In these systems, the starting material is in liquid form and solidifies through curing (by light, heat, or chemical processes).
- The following AM systems fall into this category:
 - Stereolithography Apparatus (SLA)
 - Solid Ground Curing (SGC)
 - Digital Light Processing (DLP)
 - Masked Stereolithography (MSLA)
 - Lithography-based Metal Manufacturing (LMM)

2. Solid-based Systems:

- The material is in solid form such as wire, sheets, laminates, or pellets.
- The process may involve cutting and bonding/joining or melting and fusing.
- The following AM systems fall into this definition:
 - Laminated Object Manufacturing (LOM)
 - Fused Deposition Modelling (FDM)
 - Paper Lamination Technology (PLT)
 - Multi-Jet Modelling (MJM)
 - Selective Lamination Composite Object Manufacturing (SLCOM)

3. Powder-based Systems:

- Material is in the form of fine powder particles (metals, polymers, or ceramics).
- Parts are formed by binding or fusing the powder, using a laser, electron beam, or binder solution.
- The following AM systems fall into this group:
 - Selective Laser Sintering (SLS)
 - Direct Metal Laser Sintering (DMLS)
 - Three-Dimensional Printing (3DP)
 - Laser Engineered Net Shaping (LENS)
 - Electron Beam Melting (EBM)
 - Direct Metal Deposition (DMD)

II. BASED ON PROCESS MECHANISM (ISO/ASTM Standard Classification):

1. Vat Photopolymerization:

- Process where liquid photopolymer is selectively cured by a light source.
- Examples:
 - Stereolithography (SLA)
 - Digital Light Processing (DLP)
 - Masked Stereolithography (MSLA)
 - Lithography-based Metal Manufacturing (LMM)

2. Material Extrusion:

- Material is selectively dispensed through a nozzle or orifice.
- Example:
 - Fused Deposition Modelling (FDM)

3. Powder Bed Fusion:

- Thermal energy selectively fuses regions of a powder bed.
- Examples:
 - Selective Laser Sintering (SLS)
 - Direct Metal Laser Sintering (DMLS)
 - Electron Beam Melting (EBM)
 - Multi Jet Fusion (MJF)

4. Material Jetting:

- Droplets of build material are selectively deposited.
- Examples:
 - Material Jetting (MJ)
 - Nanoparticle Jetting (NPJ)

5. Directed Energy Deposition:

- Focused thermal energy melts material as it is being deposited.
- Examples:
 - Laser Engineered Net Shaping (LENS)
 - Electron Beam Additive Manufacturing (EBAM)
 - Direct Metal Deposition (DMD)

6. Sheet Lamination:

- Sheets of material are bonded together layer by layer.
- Examples:
 - Laminated Object Manufacturing (LOM)
 - Selective Lamination Composite Object Manufacturing (SLCOM)
 - Plastic Sheet Lamination (PSL)

Summary of popular AM processes with Material used and Application

Category	Process	Material Used	Applications
<i>Liquid-based</i>	Stereolithography (SLA)	Photopolymer resin	Dental, Jewellery, Medical Devices, Prototypes
	Material Jetting (PolyJet, MJM)	Liquid photopolymers	Multi-colour prototypes, Medical models, Realistic parts
<i>Solid-based</i>	Fused Deposition Modelling (FDM)	Thermoplastic filament (PLA, ABS, Nylon)	Prototyping, Jigs & Fixtures, Low-cost models
	Laminated Object Manufacturing (LOM)	Sheets of paper, plastic	Visual models, Design studies

Powder-based	Selective Laser Sintering (SLS)	Polymer powders (Nylon, TPU)	Functional prototypes, Automotive & Aerospace parts
	Direct Metal Laser Sintering (DMLS)	Metal powders (Steel, Ti, Al, Inconel)	Aerospace, Medical implants, Tooling
	Binder Jetting	Sand / metal powders	Casting moulds, Full-colour models, Metal parts
	Laser Engineered Net Shaping (LENS)	Metal powder	Repair of Turbine blades, Large aerospace parts

1.1.5 ADDITIVE MANUFACTURING IN NEW PRODUCT DEVELOPMENT

Product is just an idea before its actual development. Product development typically refers to all stages involved in bringing a product from concept or idea through market release and beyond. In other words, the product development process incorporates a product's entire journey.



Flowchart shows the steps in New Product Development

AM supports the New Product Development cycle from idea generation to market launch by providing speed, cost savings, design freedom, and flexibility. It is especially valuable in industries requiring customization, lightweighting, and rapid innovation.

1. Idea Generation:

- The first stage of New Product Design where innovative product concepts are conceived.
- Market trends, customer needs, and technological opportunities are analysed.
- AM helps here by enabling quick visualization and even rough early prototypes to communicate ideas more effectively.

2. CAD Design:

- The product idea is converted into a 3D CAD model.
- Designers focus on form, function, and aesthetics.
- AM encourages design freedom, since constraints of traditional manufacturing (e.g., machining, casting) are reduced. Lightweight structures and internal channels can be designed without concern for tooling limitations.

3. Prototype Development:

- Turning concepts into actual product designs with engineering specifications, prototypes, and models.
- Unlike traditional methods, in AM no moulds, dies, or complex setups are required.

- AM allows low-cost prototyping, even for complex geometries.
- Functional (or) aesthetic prototypes are produced quickly

4. Testing & Validation:

- Prototypes are subjected to functional tests (strength, ergonomics, thermal, aerodynamic, etc.).
- Fit and assembly are checked with other components.
- AM enables early validation, reducing chances of late-stage design failures.
- Multiple prototype versions can be tested in parallel.

5. Design Iteration:

- Based on testing feedback, design changes are incorporated.
- AM makes rapid iteration possible: quick redesigns, reprinting, and testing.
- Iteration cycles become shorter, improving the overall design quality.

6. Final Production:

- Once optimized, the product moves to production.
- For mass production, conventional methods such as casting, injection moulding, machining may be used.
- For low-volume, complex, or customized products, AM itself can serve as the final production method (e.g., aerospace parts, dental implants).
- Hybrid approaches (AM + machining) are also sometimes used

7. Market Launch:

- Full-scale production, marketing, and distribution of the new product into the market.
- With faster prototyping and reduced design cycles, time-to-market is significantly reduced.
- Early product testing with real users becomes possible, improving acceptance.
- Companies gain a competitive advantage by launching innovative products ahead of competitors.

1.1.6 MATERIALS IN ADDITIVE MANUFACTURING

Additive Manufacturing makes use of a wide range of materials depending on the process, application, and performance requirements. These materials can be broadly classified into Polymers, Metals, Ceramics, Composites, and Biomaterials.

Each class is available in specific forms (filaments, powders, resins, sheets, etc.) and is compatible with certain Additive Manufacturing processes.

1. Polymers: Polymers are the most common entry point into AM due to low cost, ease of processing, and versatility.

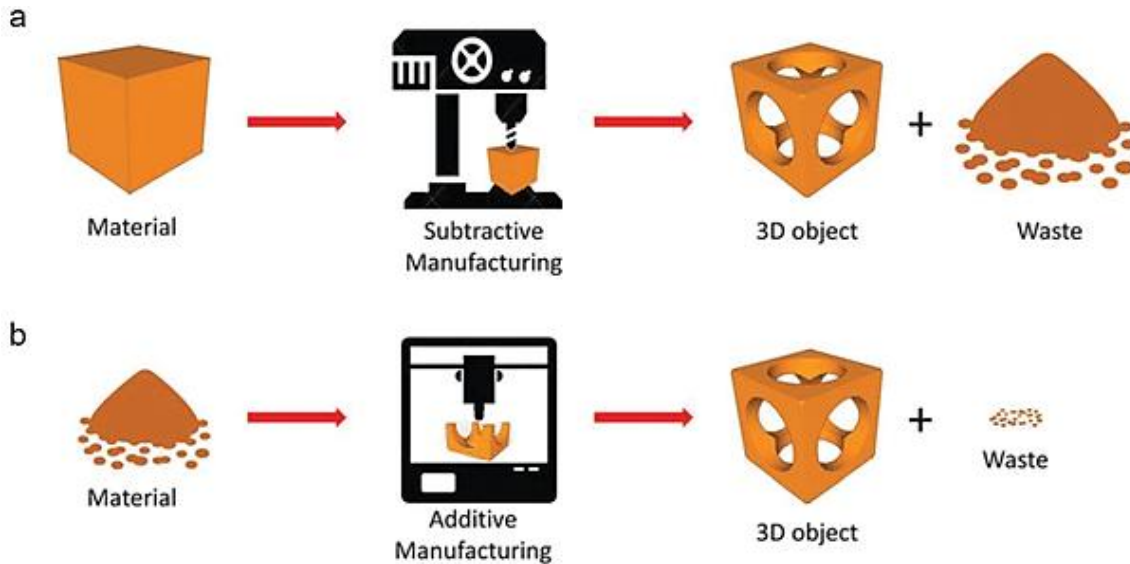
- Forms: Filament, Powder, Resin
- Processes & Materials:
 - FDM: ABS, PLA, PETG, Nylon, PEEK – widely used for ease of printing and affordability.
 - SLA/DLP: Acrylic and Epoxy based Photopolymer resins – suitable for high-resolution models with smooth surface finish.

- SLS: Nylon (PA12), Thermoplastic elastomers, Polyurethane – preferred for strong, flexible functional parts.
 - Applications: Prototypes, consumer products, medical models, functional tools, and low-volume end-use parts.
2. Metals: Metals in AM are valued for high strength, durability, and ability to replace conventionally machined parts.
- Forms: Powder, Wire
 - Processes & Materials:
 - DMLS, LENS, DMD: Stainless Steel, Aluminium, Titanium, Inconel, Co-Cr alloys.
 - WAAM: Titanium, Aluminium, Steel – cost-effective for large-scale components.
 - Applications: Aerospace (lightweight structural parts), automotive (engine components), medical implants (titanium alloys), tooling, and defence equipment.
3. Ceramics: Ceramics offer biocompatibility and excellent thermal resistance, making them essential in medical and high-heat environments.
- Forms: Powder
 - Processes & Materials:
 - Binder Jetting: Alumina, Zirconia, Silicon Carbide, Hydroxyapatite.
 - Applications: Dental crowns and bridges, orthopaedic implants, high-temperature components, electronics, and thermal barrier coatings.
4. Composites: Composites combine strength and lightweight properties, enabling high-performance applications.
- Forms: Filament, Sheet
 - Processes & Materials:
 - FDM: Polymer filaments reinforced with Carbon fiber, Glass fiber and Kevlar.
 - CBAM: Metal or ceramic matrices reinforced with fibers.
 - Applications: Lightweight structural parts in aerospace and automotive industries, sporting goods, robotics, and defence.
5. Biomaterials: Biomaterials are crucial for bioprinting and healthcare innovations, focusing on biocompatibility and tissue integration.
- Forms: Hydrogels, Biopolymers (filament & powder), Bio-inks
 - Processes & Materials:
 - SLA/DLP, SLS, FDM: Collagen, Gelatine, Alginate, Chitosan, PEG, PLA blends.
 - Applications: Tissue engineering scaffolds, artificial organs, drug delivery systems, prosthetics, and regenerative medicine.

1.1.7 ADDITIVE V/S CONVENTIONAL MANUFACTURING PROCESSES

Additive Manufacturing and Traditional (Subtractive) Manufacturing are two fundamentally different approaches to making parts. In AM, also known as 3D printing,

material is deposited layer by layer to build the product directly from a CAD model. This enables the creation of complex geometries, lightweight structures, and customized designs without the need for tooling. On the other hand, Subtractive Manufacturing is the traditional process where material is removed from a solid block through machining operations such as milling, drilling, or turning to achieve the desired shape. While AM offers design freedom and rapid prototyping, Subtractive Manufacturing provides high precision, smooth surface finish, and is well-suited for large-scale production. Together, these two approaches represent complementary methods in modern product development.



Differences between Additive and Subtractive Manufacturing

Aspect	Additive Manufacturing	Subtractive Manufacturing
Definition	Builds parts layer by layer by adding material.	Creates parts by removing material from a solid block (e.g., turning, milling, drilling).
Material Usage	Material-efficient (only deposits where needed, less waste).	High material wastage (chips, scrap from cutting).
Design Freedom	High – can produce complex geometries, lattices, internal channels.	Limited by tool access, cutting angles, and machine constraints.
Tooling Requirement	No special tools or moulds needed; directly from CAD.	Requires cutting tools, fixtures, jigs, and sometimes moulds.
Prototyping Speed	Rapid prototyping; easy design changes.	Slower prototyping; each change may need new tooling/setup.
Surface Finish	Generally rough, requires post-processing for smooth finish.	Very fine and smooth finish possible directly from machining.
Material Types	Polymers, metals, ceramics, composites	Almost all engineering materials (metals, plastics, wood, etc.).
Production Volume	Best for low-volume, customized, or complex parts.	Best for high-volume, standardized mass production.
Cost	Cost-effective for prototyping, but expensive for large batches.	Cost-effective for mass production.

1.1.8 APPLICATIONS OF ADDITIVE MANUFACTURING IN VARIOUS DOMAINS

1. Aerospace & Defence:

- Lightweight structural parts (brackets, housings).
- Complex fuel nozzles, turbine blades with internal cooling channels.
- Spare parts production on demand (reduces inventory).
e.g.: GE Aviation's 3D-printed fuel nozzle for LEAP engine.

2. Automotive:

- Rapid prototyping of car components.
- Custom tooling, jigs, and fixtures for assembly.
- Lightweight metal and plastic parts (interior panels, mounts, engine brackets).
e.g.: BMW, Ford, and Tesla use AM for prototyping and end-use parts.

3. Medical & Healthcare:

- Patient-specific implants (orthopaedic, dental, cranial).
- Customized prosthetics and orthotics.
- Surgical planning models (organ replicas, bone structures).
- Bioprinting of tissues and scaffolds (research stage).
e.g.: Titanium hip implants and dental crowns.

4. Consumer Products:

- Customized eyewear, footwear, and jewellery.
- Fashion and lifestyle products (bags, watches, apparel accessories).
- Toys, gadgets, and personalized household items.
e.g.: Adidas 4D shoes with 3D-printed midsoles.

5. Industrial & Manufacturing:

- Production of jigs, fixtures, and tooling inserts.
- Injection moulding inserts with conformal cooling channels.
- Spare parts supply chain optimization through on-demand printing.
e.g.: Siemens uses AM for turbine maintenance tools.

6. Architecture & Construction:

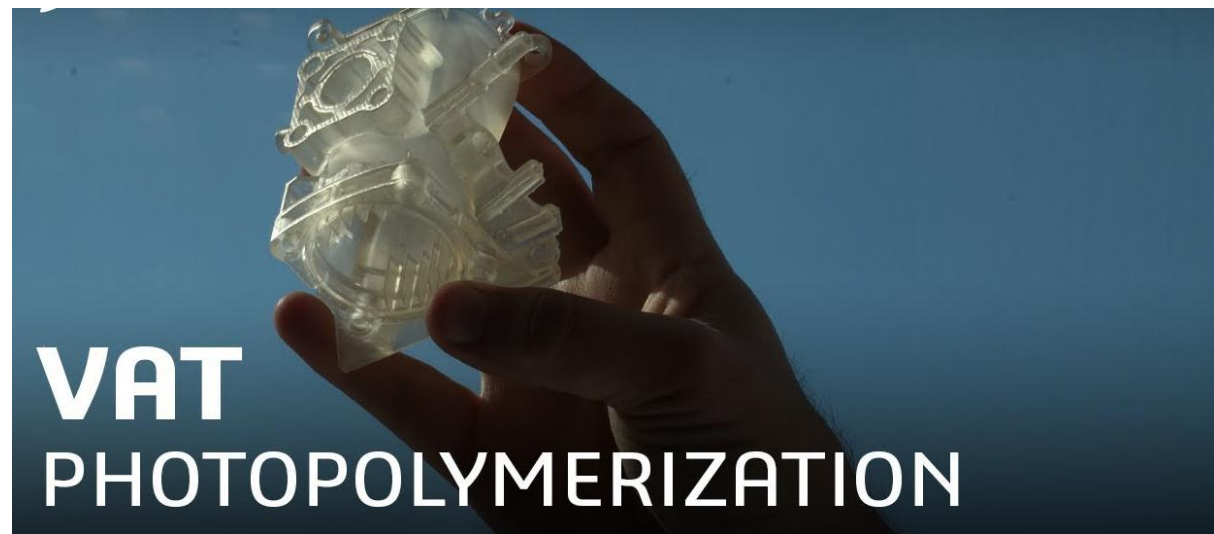
- 3D-printed houses, bridges, and structural components using concrete and composites.
- Scaled architectural models for design visualization.
- Sustainable construction using local raw materials.
e.g.: ICON's 3D-printed homes in the USA.

7. Education & Research:

- Hands-on prototyping for engineering and design students.
- Training for robotics, product design, and mechanical systems.
- Research into bioprinting, new materials, and hybrid processes.

8. Art, Jewellery & Fashion:

- Intricate jewellery designs with precious metals.
- Artistic sculptures and creative installations.
- Fashion accessories and custom designs.



COURSE CONTENT

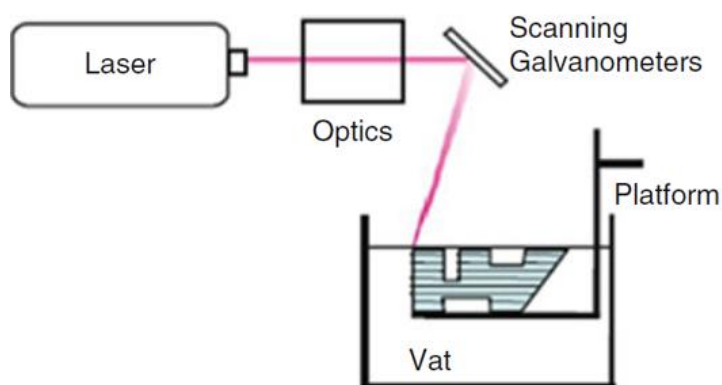
- 1.2.1 Introduction**
- 1.2.2 Stereolithography Apparatus (SLA)**
- 1.2.3 Applications of SLA**
- 1.2.4 SLA Materials**

Prof. Shiva Krishna J

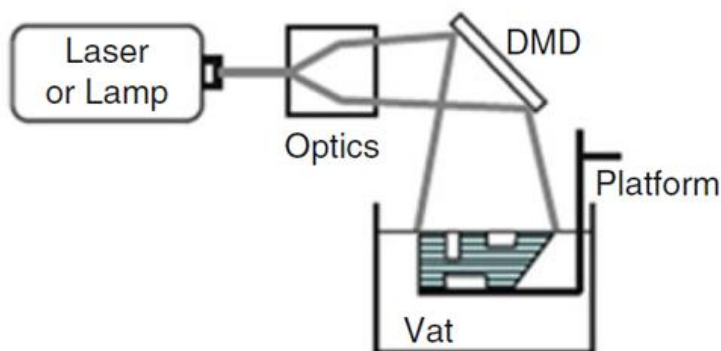
Dept. of Mechanical Engg., Agnel Institute of Technology and Design, Assagao

1.2.1 INTRODUCTION

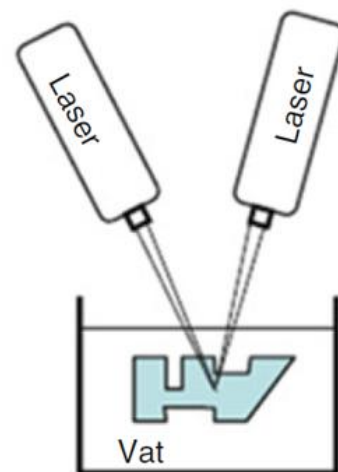
- VAT Polymerization (or) Photopolymerization processes make use of liquid, radiation-curable resins, or photopolymers, as their primary materials.
- Most photopolymers react to radiation in the ultraviolet (UV) range of wavelengths, but some visible light systems are used as well.
- Upon irradiation, these materials undergo a chemical reaction to become solid.
- The VAT Polymerization configurations include:
 1. Vector scan, or point-wise, approaches typical of commercial SL machines
 2. Mask projection, or layer-wise, approaches, that irradiate entire layers at one time,
 3. Two-photon approaches that are essentially high-resolution point-by-point approaches



Vector scan approach



Layer-wise, approach



Two-photon approach

The following process make use of VAT polymerization technique for building parts:

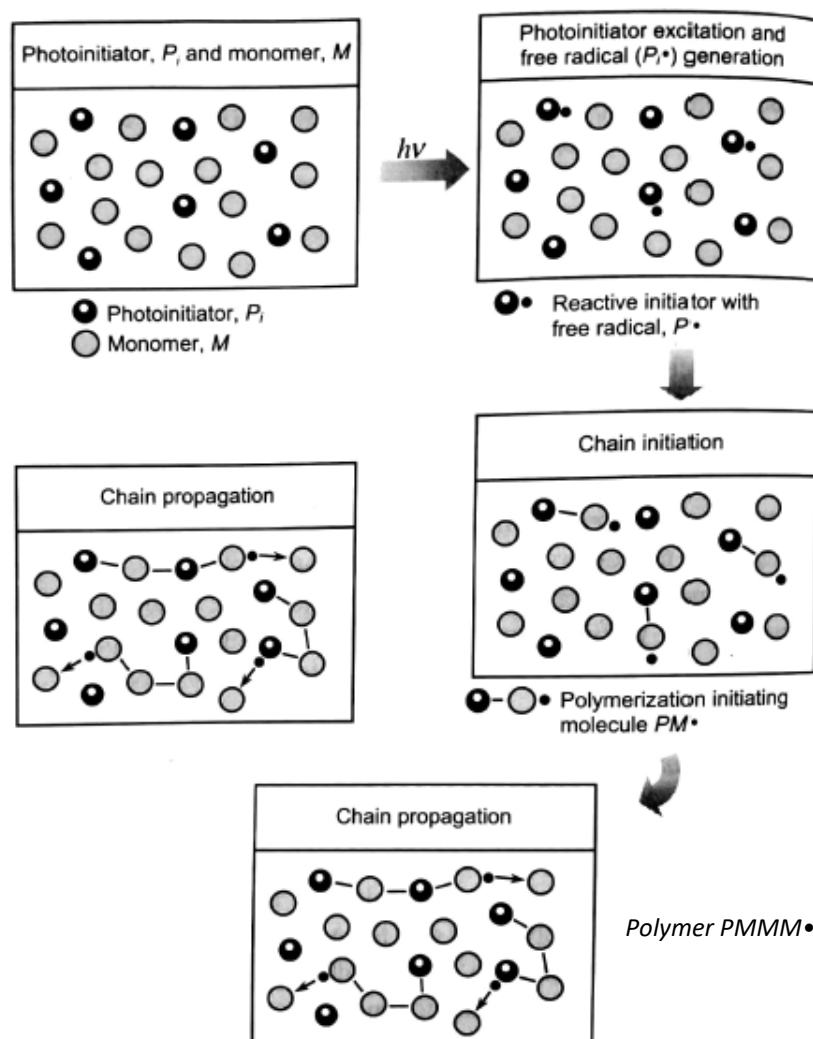
- Stereolithography Apparatus (SLA) – Uses a UV laser to selectively cure liquid photopolymer resin layer by layer.
- Digital Light Processing (DLP) – Employs a digital light projector to cure entire resin layers at once, enabling faster builds.
- Lithography-based Metal Manufacturing (LMM) – Utilizes photopolymer resin mixed with metal particles, later sintered to form dense metal parts.
- Light Enabled Additive Production (LEAP) – A high-speed process that cures resin using advanced light projection for rapid prototyping.

- Projection Micro Stereolithography (PμSL) – Provides microscale resolution using projected light patterns for fabricating microstructures.
- Masked Stereolithography (MSLA) – Uses an LCD mask with UV light to simultaneously cure entire resin layers, combining high speed and low cost.

In the present chapter will focus only on SLA process in detail with respect to its working principle, advantages, limitations, materials used and applications.

VAT POLYMERIZATION

Polymerization is the process of linking small molecules (known as monomers) into chain-like larger molecules (known as polymers). When the chain-like polymers are linked further to one another, a cross-linked polymer is said to be formed.



VAT Polymerization Principle

Photo polymerization is polymerization initiated by a photochemical process whereby the starting point is usually the induction of energy from the radiation source. Photoinitiator molecules (P_i), which are mixed with the monomers, M , are exposed to a Laser source of actinic photons, with energy of $h\nu$. The photoinitiators absorb some of the photons and are in an excited state. Some of these are converted into reactive initiator molecules ($P^\bullet$),

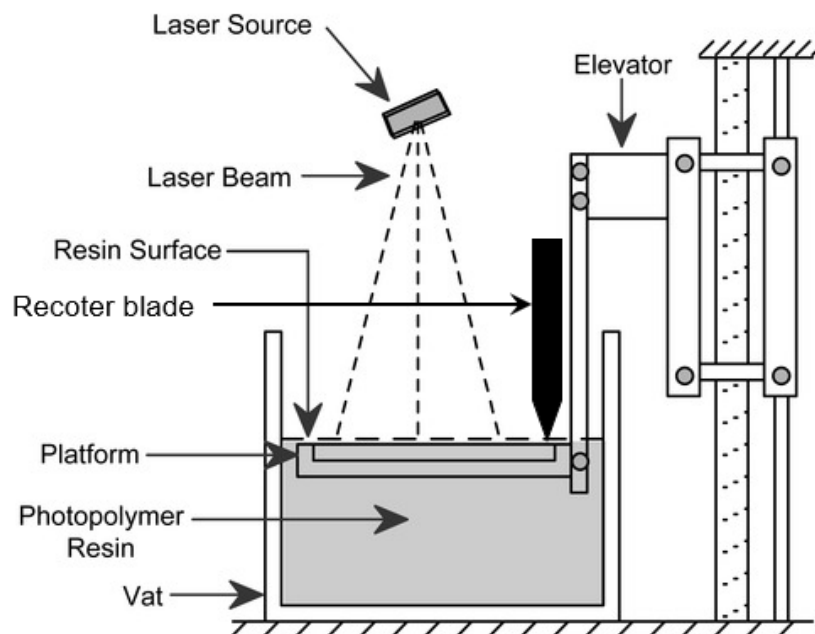
after undergoing several complex chemical energy transformation steps. These molecules then react with a monomer molecule to form a polymerization initiating molecule, (PM•). This is the chain initiation step. Once activated, additional monomer molecules go on to react in the chain propagation step, forming longer molecules, (PMMM•) until a chain inhibition process terminates the polymerization reaction. The longer the reaction is sustained, the higher will be the molecular weight of the resulting polymer.

1.2.2 STEREOLITHOGRAPHY (SLA)

3D Systems was founded in 1986 by inventor Charles W. Hull and entrepreneur Raymond S. Freed. Amongst all the commercial RP systems, the Stereolithography Apparatus (SLA) is the pioneer with its first commercial system marketed in 1988. By 2007 3D Systems is a global company that delivers advanced rapid prototyping solutions to every major market around the world.

SLA is a Liquid based RP process, parts are built from a photo-polymer resin. A photopolymer is a light-sensitive material that changes physical properties when exposed to light. Solidification happens from the impact of a laser light on the upper surface of the liquid by VAT polymerization.

HARDWARE



SLA Hardware and Principle

- Build Chamber: Contains a removable Vat that holds the build resin, a detachable, perforated build platen on a -Z axis elevator frame, and an automated resin-level checking apparatus. The build platen has a small amount of -Z movement capability, which allows the computer to maintain the exact height per layer.
- Recoater blade: A recoater blade rides along a track at the top of the vat and serves to "smooth" the liquid across the part surface to prevent a rounding of edges due to cohesion effects.

- Laser and Optics: The laser and optics required to cure the resin. The laser beam is transferred to the part surface below by a series of optics, to scan the cross section of the part being built.
- Post processing units: Include an alcohol bath and an Ultraviolet oven call the Post Curing Apparatus. Parts are washed in the alcohol or a similar solvent immediately after being removed from the machine. This step removes any extra resin that clings to the surfaces of the part. After the final supports are removed, with some build styles the parts are required to be placed in the Ultraviolet oven to finish fully curing.

PROCESS:

The process begins with the filament spools loaded into the FDM machine and the build platform positioned at the starting height. A three-dimensional CAD model, saved in STL format and sliced into layers, is loaded into the system. Any required support structures are generated to stabilize overhanging features during printing. The computer-controlled extrusion head then heats the filament to a semi-liquid state and deposits it along the tool path corresponding to the first layer on the build platform. After completing a layer, the build platform moves down by one layer height, and the extruder deposits the next layer of molten material. This layer-by-layer deposition continues from the bottom-up until the complete part is built. Once printing is complete, the part is removed from the build platform, and any support material is detached or dissolved to reveal the final component.

ADVANTAGES

1. SLA machines have **wide range of build volumes** from small to large to suit the needs of different users.
2. Parts printed **have good dimensional accuracy** within $\pm 0.1 - 0.2$ mm makes it ideal for precision prototyping.
3. SLA can obtain one of the **best surface finishes** amongst RP technologies.
4. There is a **wide range of resin materials**, from general-purpose materials to specialty materials for specific applications.
5. The process is of high resolution of 25–50 microns layer thickness allows **production of intricate geometries and build parts with complex details**.
6. SLA can produce **optically clear parts** for lenses, fluidic devices, or visual prototypes.

DISADVANTAGES

1. **Requires support structures** for printing parts that have overhangs and undercuts.
2. **Requires post-processing** for removal of supports and cleaning in solvents, which is tedious, time-consuming, and may damage the model.
3. **Requires post-curing** under UV light to achieve full strength and stability of the printed part.
4. Resin materials are generally **brittle and not suitable for high load-bearing** applications.
5. **Operational costs are relatively high** due to expensive resins, resin tanks, and maintenance.
6. **Build speed is slower** compared to some other AM processes, especially for large parts.

1.2.3 APPLICATIONS OF SLA

SLA is one of the most precise additive manufacturing technologies, well known for its ability to produce high-resolution parts with smooth surface finish using liquid photopolymer resins. Its unique features such as excellent accuracy, fine detailing, and ability to create complex geometries make it highly suitable for applications where precision and aesthetics are of prime importance. Following are the potential areas of application:

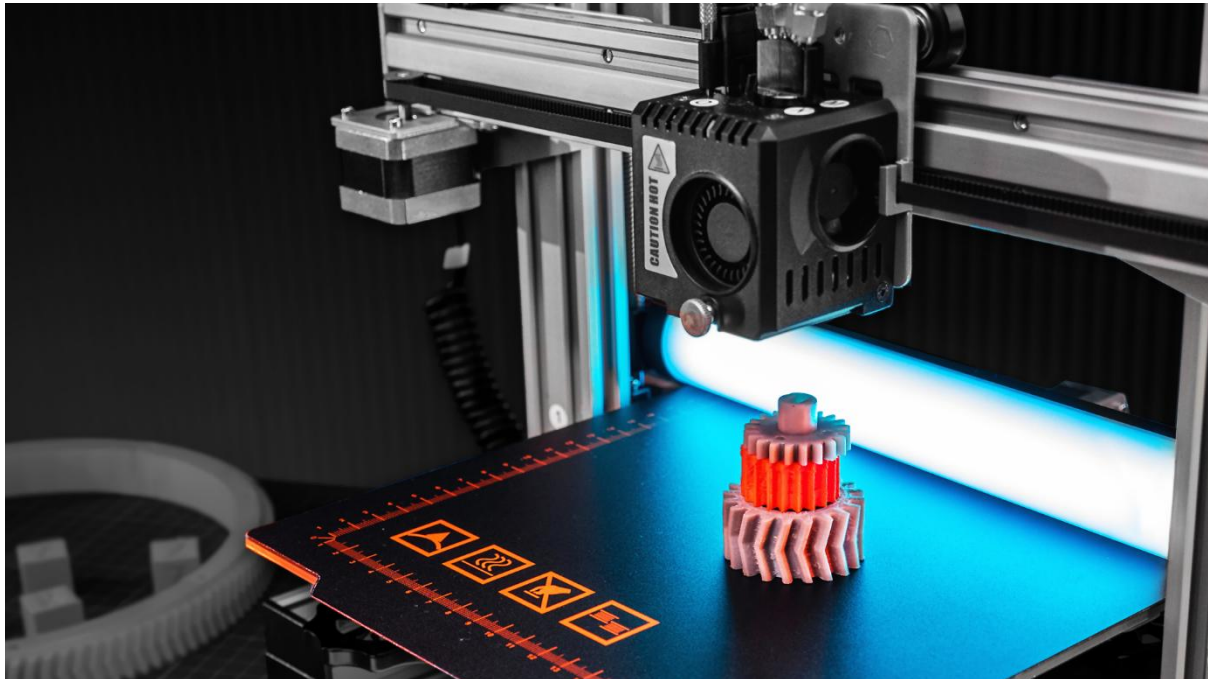
- **Concept Models** - Realistic appearance prototypes for product design validation.
- **Dental & Medical Models** - Crowns, surgical guides, anatomical models.
- **Jewellery Prototyping** - Master patterns for investment casting.
- **Miniatures & Figurines** - High-resolution prototypes with excellent aesthetics.
- **Optics Prototyping** - Lenses, casings, and light guides.

1.2.4 SLA MATERIALS

SLA resins are generally UV-curable photopolymers, based on acrylates, epoxies, urethanes, or composites with fillers (glass, ceramic, wax, carbon). Each formulation tailors properties for specific applications.

SLA Materials – Composition, Properties, and Applications

Category	Material Type	Key Properties	Typical Applications
Standard Resins	Photopolymer acrylates/epoxies	Smooth surface, high detail, brittle under stress	Concept models, visual prototypes
Engineering Resins	Urethane-based resins	Impact-resistant, ABS/PP-like strength	Functional prototypes, snap-fits
	Glass/ceramic-filled photopolymers	High stiffness, dimensional stability	Jigs, fixtures, precision parts
	Heat-resistant photopolymers	Withstands high heat, thermal stability	Mold masters, heat-resistant parts
	Urethane/elastic photopolymers	Rubber-like, bendable, resilient	Gaskets, seals, wearables
Specialty Resins	Wax-filled or castable acrylates	Clean burnout, no ash residue	Jewelry, dental crowns, investment casting
	Clear methacrylate resins	Optical clarity, smooth finish	Lenses, light pipes, fluidic devices
	Biocompatible methacrylate/epoxy blends	Safe for medical/dental use, sterilizable	Surgical guides, prosthetics
	Ceramic-filled resins	Very rigid, good thermal and dimensional stability	Industrial models, engineering prototypes
Dental/Medical Resins	Methacrylate-based	High precision, tooth-colored	Dental aligners, models
	Biocompatible methacrylate resins	Sterilizable, safe for patient contact	Surgical navigation aids



COURSE CONTENT

- 2.1.1 Introduction**
- 2.1.2 Fused Deposition Modelling (FDM)**
- 1.3.3 Applications of FDM**
- 1.3.4 FDM Materials**
- 1.3.5 Process Parameters**

Prof. Shiva Krishna J

Dept. of Mechanical Engg., Agnel Institute of Technology and Design, Assagao

1.3.1 INTRODUCTION

Solid-based processes use material in solid form, such as filaments (wires), sheets, laminates, or pellets, as the primary feedstock. The fabrication of parts is achieved either by melting and extruding the material or by cutting, stacking, and bonding sheets into successive layers that follow the digital model's geometry. These methods rely on thermomechanical transformation, adhesive bonding, or lamination techniques to create functional parts.

Compared to liquid-based photopolymerization processes, solid-based processes are particularly well-suited for thermoplastics, composites, and laminated structures, offering versatility in both rapid prototyping and functional end-use components. Their advantages include material availability, cost-effectiveness, and suitability for large-scale builds.

Solid-based processes can be categorized into two main groups:

1. Extrusion-Based Processes: In extrusion-based methods, a thermoplastic or composite material in filament, wire, or pellet form is heated, melted, and selectively deposited through a nozzle to build the part layer by layer. The following processes build parts using extrusion:
 - Fused Deposition Modeling (FDM) – Uses thermoplastic filaments extruded through a heated nozzle to create parts.
 - Multi-Jet Modeling (MJM) – Jets multiple streams of molten material often thermoplastic or wax to build detailed parts layer by layer.
2. Lamination-Based Processes: In lamination-based methods, sheets or laminates of material (such as paper, plastic, or composites) are cut to the required shape and bonded together to form the part.
 - Laminated Object Manufacturing (LOM) – Involves cutting and bonding layers of sheet material (paper or plastic foil) with heat and adhesive.
 - Paper Lamination Technology (PLT) – Uses layers of standard or coated paper bonded with adhesive and cut with a laser to form 3D objects.
 - Selective Lamination Composite Object Manufacturing (SLCOM) – Employs composite sheets with fiber reinforcement, laminated and cut to produce high-strength functional parts.
 - Composite Based Additive Manufacturing (CBAM) – Uses sheets of dry fiber reinforcement (carbon or fiberglass), bonded with a polymer matrix and cut to shape, creating lightweight and strong composite structures.

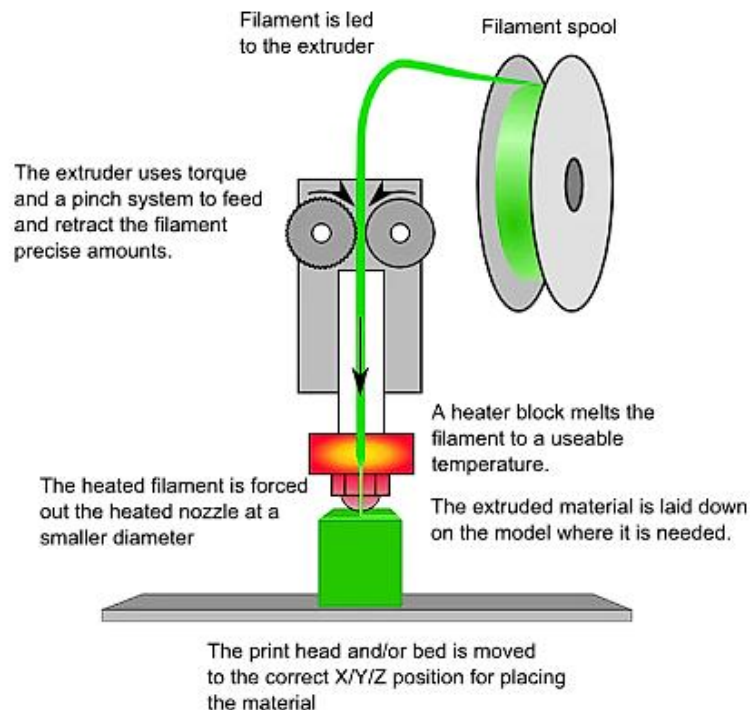
In the present chapter, we will focus specifically on the FDM process, as it is the most widely adopted extrusion-based solid AM technique

1.3.2 FUSED DEPOSITION MODELLING (FDM)

Stratasys Inc., founded in 1989, has played a pioneering role in developing products based on FDM technology. The process was first invented by Scott Crump in 1988, with the patent officially granted in the United States in 1992.

FDM is an extrusion-based rapid prototyping process, although it works on the same layer-by-layer principle as other RP systems. Also known as Fused Filament Fabrication

(FFF). It is one of the most widely used additive manufacturing processes due to its simplicity, cost-effectiveness, and ability to fabricate functional thermoplastic parts.

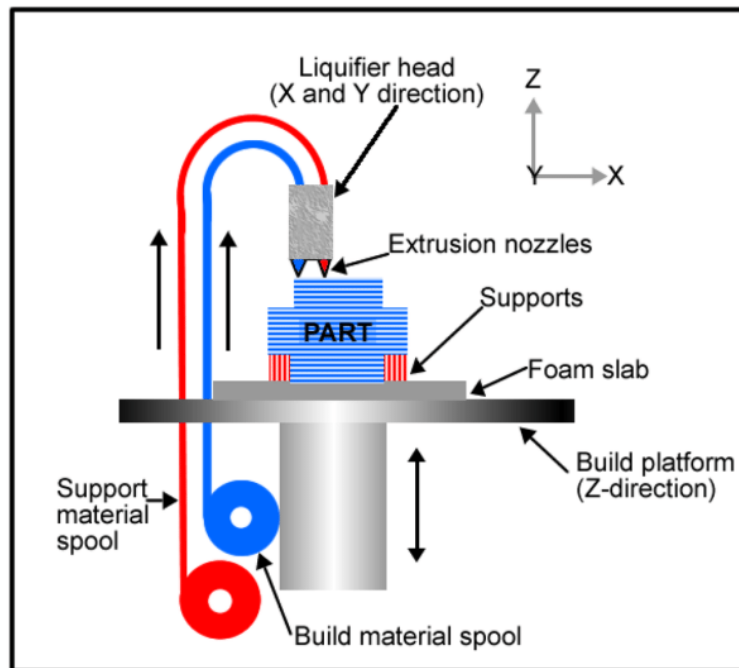


Extrusion principle in FDM

The principle of FDM is based on surface chemistry, thermal energy, and layer manufacturing technology. In this process, a thermoplastic material in the form of a continuous filament spool is fed into a specially designed extrusion head. Inside the head, the filament is heated until it softens or melts, and then it is extruded through a fine nozzle. As the molten material is deposited, it undergoes cooling and solidification, bonding with the previously deposited layer. By repeating this layer-by-layer deposition and solidification cycle, the complete three-dimensional part is fabricated, similar to other AM systems.

HARDWARE

- **Build Platform**: It is the platform on which the part is built layer by layer. It moves in – Z direction.
- **Material spools**: The modeling material is in spools - very much like a fishing line. Both build material and support material are supplied in spool form. The filament on the spools is fed into an extrusion head.
- **Extrusion head**: It heats the material to a semi-liquid state. Two parallel wheels attached to a small electric motor by gears grip the material and feed it to the heaters. The heating element is electronically controlled and has feedback thermocouples to allow for a stable temperature throughout.
- **Extrusion nozzles**: Two Extrusion nozzles (or) tips are externally threaded and screw up into the heating chamber exit. The tips can be removed and replaced with different size openings.



Hardware and Principle of FDM

PROCESS

A geometric model of the part is created using CAD software and saved in STL format. The STL file is sliced into horizontal layers, taking into account part orientation and any necessary support structures. Slice thickness can be adjusted between 0.10 mm to 0.50 mm depending on desired resolution and surface finish. The filament material from the spools is fed into the extrusion head and heated to a semi-liquid state. Dual extrusion systems allow one nozzle to print the primary build material and the other to deposit support material. The molten filament is extruded through the nozzle and deposited as ultrathin layers onto the build platform. The surrounding air is kept below the filament's melting point, allowing the material to solidify rapidly after deposition. The part is built layer by layer, from the bottom up, with each new layer bonding to the previous one through thermal fusion.

ADVANTAGES:

1. **Variety of Material Choice:** Material used for printing is available in various options. Some common of them are PLA, PETG, ABS, PET, TPU and Nylon.
2. **Compact Design:** FDM technology-based printers are quite simple and thus do not take up too much space. The size of the printer is totally dependent on the build volume. And other parts like the outer frame, power unit, etc. don't take much space.
3. **Minimal wastage:** The FDM process builds parts directly by extruding semi-liquid melt onto the model. Thus, only those material needed to build the part and its support are needed, and material wastages are kept to a minimum.
4. **Ease of support removal:** With the use of Break Away Support System (BASS) and Water-Soluble Support System, support structures generated during the FDM building process can be easily broken off or simply washed away.
5. **Ease of material change:** Build materials, supplied in spool form are easy to handle and can be changed readily. This keeps the operation of the machine simple and the maintenance relatively easy.

LIMITATIONS:

1. **Restricted Accuracy:** The diameter of standard filaments (commonly 1.75 mm or 2.85 mm) limits the achievable dimensional accuracy of FDM parts.
2. **Slower Build Speed:** Printing large cross-sectional areas can be time-consuming, as deposition rate is limited by the extrusion speed of the material.
3. **Unpredictable Shrinkage and Warping:** Rapid cooling of extruded material induces internal stresses, which can cause shrinkage, warping, or dimensional distortions in the final part.
4. **Surface Finish Limitations:** FDM-printed parts generally have visible layer lines and rough surfaces, requiring post-processing such as sanding, priming, or coating to achieve smooth finishes.

1.3.3 APPLICATIONS OF FDM

FDM is one of the most widely used additive manufacturing technologies, known for its ability to build durable parts using thermoplastic materials in filament form. Its unique features such as cost-effectiveness, ease of use, ability to process a wide range of engineering-grade polymers, and suitability for functional prototyping make it a versatile AM technique. FDM is especially valued for producing parts with good mechanical strength, material stability, and reasonable accuracy, making it a preferred choice for both prototypes and end-use applications. Following are the potential areas of application:

- **Functional Prototypes** - Fit, form, and functional testing of parts.
- **End-Use Parts** - Low-volume production of customized components and replacement parts.
- **Tooling & Fixtures** - Jigs, fixtures, and assembly tools for manufacturing.
- **Automotive & Aerospace** - Strong, lightweight thermoplastic parts for testing or end-use.
- **Consumer Products** - Low-cost customized parts, housings, enclosures.
- **Educational Use** - Affordable 3D printing for teaching and training.

1.3.4 FDM MATERIALS

FDM filaments are generally thermoplastic polymers supplied in spool form, including materials such as PLA, ABS, PETG, Nylon, TPU, or composites reinforced with carbon fiber, glass fiber. Each filament formulation is designed to provide specific mechanical, thermal, or functional properties, making it suitable for applications ranging from concept models and prototypes to functional and end-use parts.

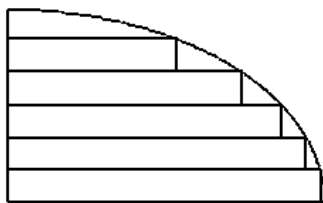
FDM Materials – Composition, Properties, and Applications

Category	Material Type	Key Properties	Typical Applications
Standard Filaments	Polylactic Acid (PLA)	Easy to print, low warping, biodegradable	Concept models, educational models, visual prototypes
	Acrylonitrile Butadiene Styrene (ABS)	Strong, durable, heat-resistant	Functional prototypes, automotive parts, enclosures

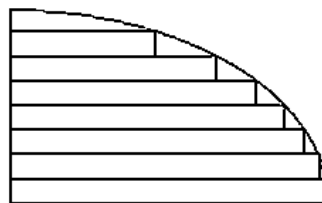
	Polyethylene Terephthalate Glycol (PETG)	Flexible, chemically resistant, moderate strength	Mechanical parts, housings, protective cases
Engineering Filaments	Nylon (Polyamide)	High strength, wear-resistant, flexible	Functional prototypes, gears, bearings, load-bearing parts
	Polycarbonate (PC)	Tough, impact-resistant, heat-stable	Engineering components, automotive, electrical housings
	Polyether ether ketone (PEEK)	High strength, high-temperature resistance, chemical resistant	Aerospace, medical implants, industrial applications
Composite Filaments	Carbon Fiber Reinforced PLA, Nylon, PETG	High stiffness, dimensional stability, lightweight	Structural parts, jigs, fixtures, drones
	Glass Fiber Reinforced PLA, Nylon, PETG	Increased strength, heat resistance	Functional parts requiring durability and stability
Flexible / Specialty Filaments	Thermoplastic polyurethanes (TPU) & Thermoplastic Elastomers (TPE)	Rubber-like, flexible, resilient	Gaskets, seals, wearable devices

1.3.5 PROCESS PARAMETERS

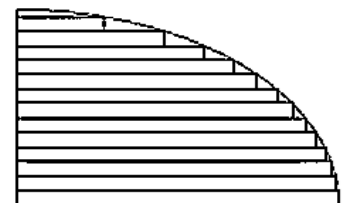
1. Layer thickness:



Thick layer
Rough surface
Less building time



Middle layer thickness
Smooth surface
Moderate Building time less



Very thin layer
Very smooth surface
High building time

Effect of layer thickness on surface finish and part building time

Layer thickness refers to the thickness of the bead extruded from the nozzle tip or the vertical distance (Z-direction) between successive layers. It directly affects build time, surface quality, and resolution, particularly on inclined surfaces. The achievable layer thickness depends on the machine specifications, nozzle diameter, and type of filament material used.

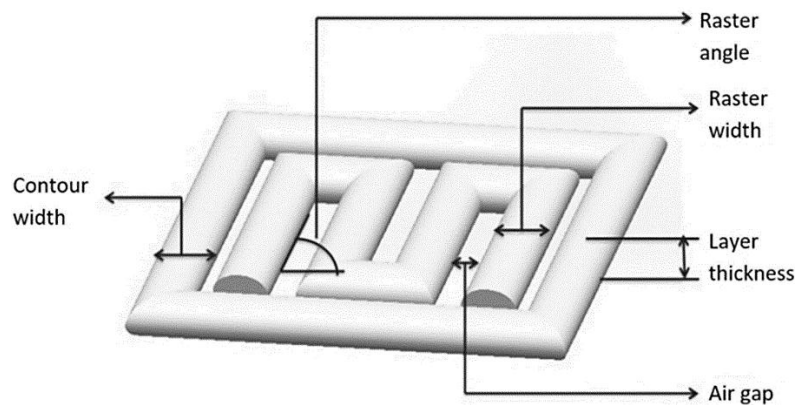
2. Bed Temperature:

Bed temperature influences the adhesion of the part to the build platform. Optimal adhesion is achieved when the bed is slightly above the Glass transition Temperature of the material, as this increases the contact area due to thermal expansion and reduces warping. Excessively high bed temperatures, however, may lead to distortion or sagging of the part.

3. Extruder Temperature:

The extruder temperature controls the viscosity and flow of the molten filament, affecting both interlayer bonding and the surface finish of the part. Higher temperatures generally improve adhesion to the previous layer and produce smoother surfaces, but excessively high temperatures can cause material deformation or stringing. Most FDM machines automatically adjust the extruder temperature based on the material type and maximum allowable limits.

4. Raster and Contour width:

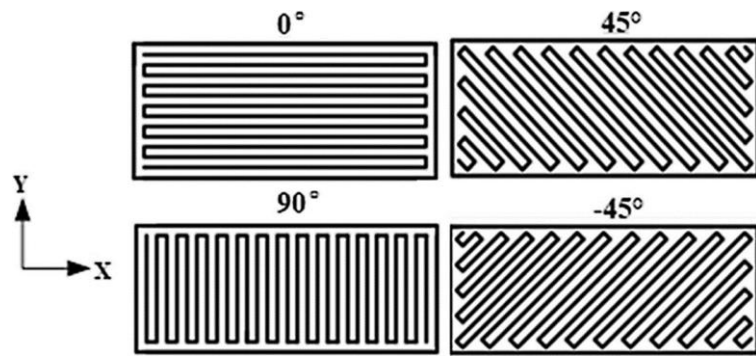


Raster width, Contour width and Raster Angle

- **Raster Width:** This is the width of the extruded bead deposited on the build platform. It is typically 1.2–1.5 times the nozzle diameter and determines the density and strength of the internal infill.
- **Contour Width:** This refers to the width of the outermost perimeter or top layer of the part. A larger contour width is preferred for better surface finish and dimensional accuracy, while thin contours may deform during extrusion. Wider contours are also ideal for hollow or shell structures.

5. Raster Angle:

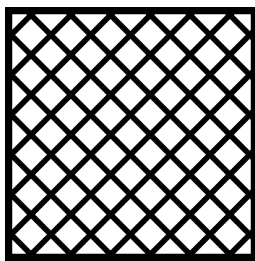
Raster angle is the angle made by the raster tool path with respect to the X-axis of the build plate. The commonly used raster angles are cross ($0^\circ/90^\circ$) and crisscross ($-45^\circ/45^\circ$). The raster angle plays a significant role in the bonding and strength of the parts. In addition, it also influences the surface finish of the built part. The crisscross pattern generally provides better surface finish.



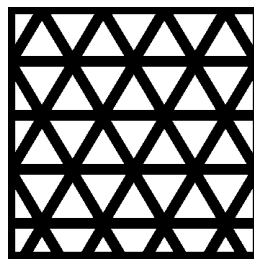
Line Infill pattern with raster angles ($0^\circ/90^\circ$) and ($-45^\circ/45^\circ$)

6. Infill Pattern:

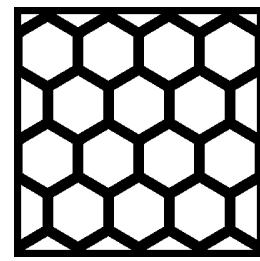
Infill pattern is the structure and shape of the material inside of a part. These can affect a part's strength, weight, print time, and even flexibility. Commonly used infill patterns are:



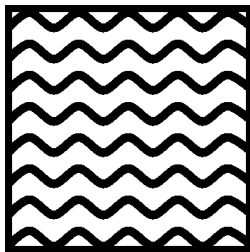
Grid



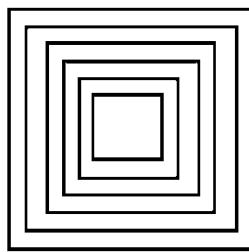
Triangle



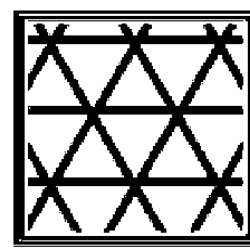
Honeycomb



Gyroid



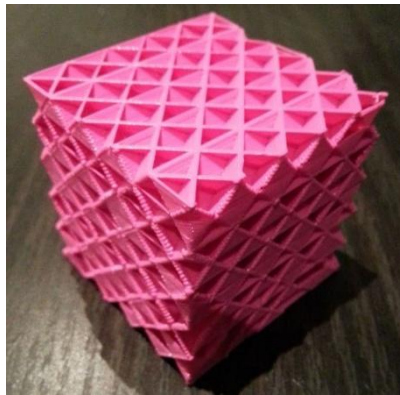
Concentric



Tetrahedral

- **Lines:** The lines infill pattern contains lines printed in one direction (either along the X or Y-axis) every other layer. This infill pattern provides strength in only two dimensions and is good for quick prints. The lines pattern doesn't use too much material and keeps weight pretty light.
- **Grid:** The grid infill pattern is similar in look to lines, but instead of one-directional lines every other layer, it contains two-dimensional lines every layer, with twice as much space in between lines. This pattern provides two-dimensional strength but is still somewhat strong. The grid pattern consumes an average amount of material to complete.
- **Triangles:** The triangles pattern looks like overlapping triangular lines, with lines going in three directions in the XY-plane. This infill pattern provides strength only in two dimensions but still works for prints that need to be strong.

- **Honeycomb:** This pattern produces a honeycomb structure, making for an appealing visual. This infill pattern is good for semi-fast prints that require moderate strength, and it doesn't consume too much material.
- **Gyroid:** The gyroid infill pattern may perhaps look the coolest but it's also arguably the strangest infill pattern. It includes concaving irregular curvatures that eventually cross paths. It's meant to strike an optimal balance between strength, material, and print time.
- **Concentric:** The concentric infill pattern is an internal structure composed of concentric lines that match a part's outline. This pattern is quick to print, good for flexible parts, and consumes significantly less material than most patterns.
- **Tetrahedral:** The tetrahedral infill pattern is very strong 3D structure, lightweight, distributes forces evenly. Used commonly for Aerospace-grade parts, high-strength lightweight applications.



Tetrahedral Infill pattern in 3D